

[This question paper contains 7 printed pages]

Your Roll No. :
 Sl. No. of Q. Paper : 7407 J
 Unique Paper Code : 32341502
 Name of the Course : B.Sc.(Hons.) Computer Science
 Name of the Paper : Theory of Computation
 Semester : V

Time : 3 Hours Maximum Marks : 75

Instructions for Candidates :

- Write your Roll No. on the top immediately on receipt of this question paper.
- All questions in **Section-A** are compulsory.
- Attempt any **four** questions from **Section-B**.
- Parts of a question must be answered together.
- Assume alphabet $\Sigma = \{a,b\}$ unless stated otherwise.

P.T.O.

Section - A

1. (a) Do the following regular expressions represent the same language (give reason) :

2

$$R_1 = ((a + b)(a + b))^* a$$

$$R_2 = (a + b)((a + b)(a + b))^* a$$

- (b) Write a regular expression and build a deterministic finite automata for the language containing all strings having **a** at every odd position.

2+3

- (c) Describe in English the languages represented by the following regular expressions :

2+2

(i) $b^*ab^*ab^*ab^* + b^*ab^*ab^*$

(ii) $(a + b)^*aa(a + b)^*$

- (d) Describe pumping lemma for regular languages.

2

- (e) Based on the language $S = \{aa, ba, ab, bb\}$, describe the language S^* .

2

(f) What are the languages generated by the following grammars : 3+3

(i) $S \rightarrow XA$
 $X \rightarrow aXb \mid \Lambda$

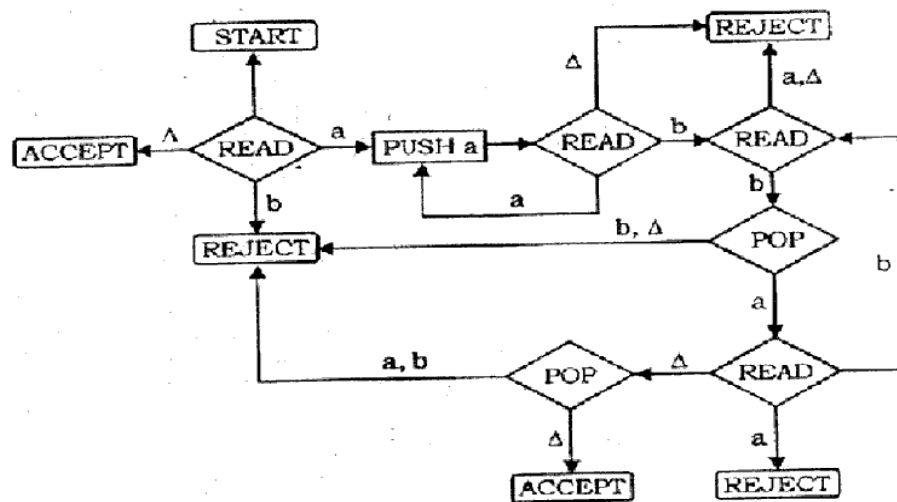
$A \rightarrow aA \mid \Lambda$

(ii) $S \rightarrow AB$
 $A \rightarrow aA \mid \Lambda$
 $B \rightarrow bB \mid \Lambda$

(g) Show that the following CFG is ambiguous : 3

$S \rightarrow X a X$
 $X \rightarrow a X \mid b X \mid \Lambda$

(h) Describe the language (in English) accepted by the following PDA : 3



(i) Describe the halting problem. 3

(j) Let $M = (K, \Sigma, \delta, s, \{h\})$, where 5

$K = \{q_0, q_1, h\}$

$\Sigma = \{a, b, \sqcup, \triangleright\}$

$S = q_0$

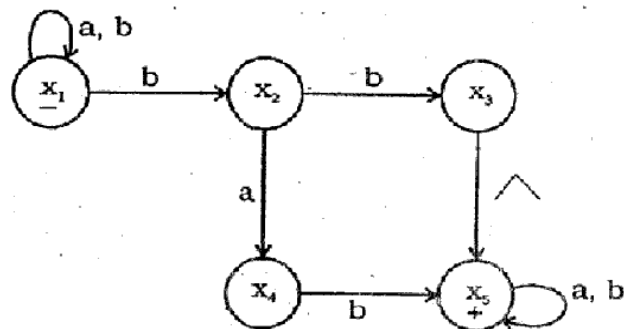
and δ is given by the following table :

q	σ	$\delta(q, \sigma)$
q_0	a	(q_1, b)
q_0	b	(q_1, a)
q_0	$\sqcup$	$(h, \sqcup)$
q_0	$\triangleright$	$(q_0, \rightarrow)$
q_1	a	$(q_0, \rightarrow)$
q_1	b	$(q_0, \rightarrow)$
q_1	$\sqcup$	$(q_0, \rightarrow)$
q_1	$\triangleright$	$(q_1, \rightarrow)$

Trace the computation of M starting from the configuration $(q_0, \triangleright aabbba)$.

Section -B

2. (i) Let L = All strings that end with aa or bb . Construct DFAs for L and L' (i.e., Complement of L). 6
- (ii) Is the language $\{a^m b^m ; m \geq 0\}$ regular? Justify using Pumping Theorem. 4
3. (i) Build an FA accepting the language comprising of all strings having first two characters same as the last two. 6
- (ii) Convert the following Transition Graph (TG) into regular expression using Bypass algorithm. 4



4. (i) Give Context Free Grammar (CFG) for the following language : 5
 $\{a^i b^j c^k \mid i + j = k; i, j, k \geq 0\}$ and $\Sigma = \{a, b, c\}$
- (ii) Build pushdown automation (PDA) to accept the following language : 5
 $\{Sb^{n+1} ; S \text{ is a string of only } a\text{'s}, n = \text{length}(S), n \geq 1\}$
5. (i) Convert the following CFG to CNF : 5
 $E \rightarrow E + E$
 $E \rightarrow E * E$
 $E \rightarrow (E)$
 $E \rightarrow 6 \mid 7$
 The terminals here are $+ * () 6 7$.
- (ii) Prove that the recursive languages are closed under complementation. 5
6. (i) Design a Turing Machine that scans to the right until it finds two consecutive a 's and then halts. The alphabet of the Turing machine should be $\{a, b, \sqcup, \Delta\}$. 5
- (ii) Prove that context-free languages closed under Union and concatenation. 5

7. (i) Build FA for each of the following regular languages L_1 and L_2 . 4
 $L_1 = b(a+b)^*$ $L_2 = a(a+b)^*b + b(a+b)^*a$
- (ii) Build FA for $L_1 \cap L_2$. 4
- (iii) Describe in English the language represented by $L_1 \cap L_2$. 2